

$$C_t \approx S_t - Ke^{-r(T-t)}$$

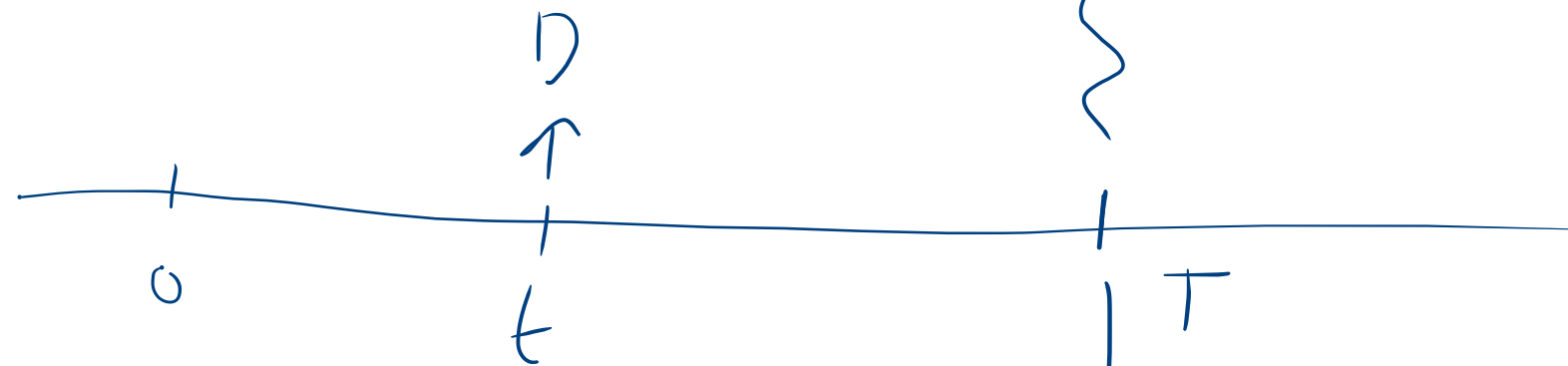
$$\theta_t = -Kne^{-r(T-t)}$$

$$\Delta_t = \frac{\partial C_t}{\partial S_t}$$

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

$$S_{t+\Delta t} = S_t e^{\underbrace{\left(\mu - \frac{\sigma^2}{2}\right) \Delta t}_A + \underbrace{\sigma (W_{t+\Delta t} - W_t)}_{\substack{B \\ \sqrt{\Delta t} \varepsilon}} \quad \text{or} \quad \varepsilon \sim N(0, 1)$$

Exo 1



$$S_0 = 22$$

$$T = 0,5$$

$$r(T) = 1,25\%$$

$$S_0 e^{r(T)T} - D e^{r_{\text{fwd}}(T-t)}$$

$$r_{\text{fwd}} \text{ t.g. } e^{r(t)t} e^{r_{\text{fwd}}(T-t)} = e^{r(T)T}$$

Exo 2

$$\text{Call deep ITM} : C_0 \approx S_0 - K e^{-rT}$$

Exo 3

$$S_{t+\Delta t} = S_t e^{(\mu - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t} \varepsilon}$$

$$\text{avec } \varepsilon \sim \mathcal{N}(0, 1)$$

$$P\left(\frac{S_{t+\Delta t}}{S_t} < 1 - \alpha\right) = P\left(\varepsilon < \underbrace{\frac{\ln(1 - \alpha) - (\mu - \frac{\sigma^2}{2})\Delta t}{\sigma\sqrt{\Delta t}}}_{d}\right)$$

Exo 4

Call/put ATM forward

$$K = S_0 e^{rT}$$

$$C_0 = P_0 \approx \underset{\substack{\uparrow \\ 1/\sqrt{2\pi}}}{0.4} S_0 \sigma \sqrt{T}$$

$$Str_0 \approx 0,8 \underset{\substack{\uparrow \\ 10}}{S_0} \sigma \sqrt{T} \underset{0,25}{=} 1 \text{€}$$

$$0,8 \times 10 \times \sigma \times 0,5 = 1$$

$$4\sigma = 1$$

$$\sigma = 25\%$$

$$Vega \approx 0,8 \times 10 \times 1\% \times 0,5 = 0,04 \text{€}$$

